

TEZPUR UNIVERSITY
Spring Semester Midterm Exam, 2022
Graph Theory (CO 217)

Maximum Marks : 40

Duration : 2 hours

You may answer as many questions as you like but maximum you get is 40.

1. Verify the following statements as true or false .

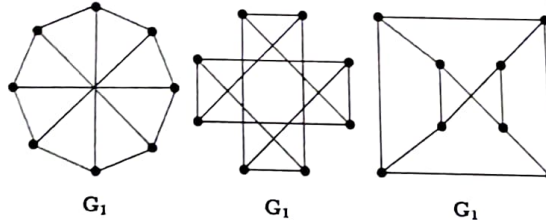
[5 × 3]

- i. Any tree T has at most one perfect matching.
- ii. Two walk between a pair of vertices p, q in a graph G implies that G has a cycle.
- iii. A bipartite graph G has at least one cycle.
- iv. Any 2-regular bipartite graph of n vertices has a vertex cover of size $\frac{n}{2}$.
- v. A graph with minimum degree 2 must have a cycle of length 3.

2. Fill in the blanks.

- i. The maximum number of edges in a connected simple graph G of n vertices is where all the vertices in G are of degree 2. [1]
- ii. A tree T has one vertex of degree 4 and one vertex of degree 3. Then T has atleast leaf nodes. [2]
- iii. The number of C_3 in $K_{3,5}$ is [2]
- iv. The number of maximum length paths in the complete graph K_n is [2]

3. Which of the following graphs are isomorphic to each other? Justify your answer. Label the vertices and show the mapping between the vertices of the isomorphic graphs. [5]



4. Prove or disprove: If G_1 and G_2 are two connected subgraphs of G having at least one vertex in common, then $G_1 \cup G_2$ is connected. [7]

5. Draw a k -regular simple graph G that has no perfect matching. Consider $k > 2$ and G to be connected. Hint: Use Tutte's theorem. [7]

6. Consider a bipartite graph G whose partition is (A, B) . Every vertex in G is of degree k . Show that $|A| = |B|$. [8]

7. If M is a maximal matching in a graph G , then $\alpha(G) \leq 2|M|$ where $\alpha(G)$ denotes the size of minimum vertex cover. [5]

